

Aliah University

BCA 4th - Semester Examination – 2025

Paper Name: Web Technology

Paper Code: CSAUGPC19

Full Marks: 80

Time: 3 hrs

Group A — (Answer all five questions) [$5 \times 2 = 10$]

1. Which of the following HTML tags is used to display a table in a web page?
CO1, PO1
2. What does CSS stand for? CO1, PO1
3. Which attribute is used to specify the URL in the <a> tag in HTML? CO1, PO1
4. Which method is used to send data to the server in the HTTP protocol? CO2, PO2
5. In JavaScript, which event is triggered when a user clicks on a button? CO2, PO2

Group B — (Answer any six questions) [$6 \times 5 = 30$]

6. Discuss the difference between inline, internal, and external CSS styles with their advantages and disadvantages. CO2, PO2
7. Examine the different types of selectors in CSS. Provide examples for each and explain their uses. CO2, PO2
8. What is the purpose of the z-index property in CSS? How is it used to manage stacking context? CO2, PO2
9. Explain the concept of AJAX in web development. How does it enhance user experience in web applications? CO3, PO3
10. Describe the key features and advantages of using the HTML5 <canvas> element. Provide an example use case. CO3, PO3
11. Compare and contrast synchronous and asynchronous JavaScript programming. How do they impact performance and user experience? CO3, PO3
12. What is the role of cookies and sessions in web development? Compare their functionality and typical use cases. CO4, PO4
13. Critically assess the significance of Responsive Web Design (RWD) in modern web development. How does the use of media queries contribute to RWD? CO4, PO4

Group C — (Answer any four questions) [4 × 10 = 40]

- ✓ 14. Analyze the importance of semantic HTML in improving website accessibility.
✓ Critically discuss the role of meta tags in HTML and their impact on SEO.
CO4, PO4
- ✓ 15. Analyze the CSS specificity hierarchy and its implications for styling conflicts.
✓ Critically discuss the concept of CSS preprocessors and their benefits in web development.
CO4, PO4
- ✓ 16. Critically analyze the pros and cons of using Bootstrap for web development.
✓ Discuss the significance of Bootstrap components in streamlining UI development.
CO4, PO4
- ✓ 17. Compare and contrast the characteristics of dynamically-typed languages like JavaScript with statically-typed languages.
✓ Evaluate the advantages and disadvantages of asynchronous programming in JavaScript, highlighting use cases.
CO5, PO5
18. Discuss the concept of the Virtual DOM and how it optimizes real DOM manipulation in modern JavaScript frameworks.
Evaluate the advantages of using frameworks like React or Vue for handling event delegation efficiently.
CO5, PO5
19. Analyze the advantages and disadvantages of using jQuery in modern web development.
Discuss the impact of AJAX in jQuery and its role in asynchronous communication on the web.
CO5, PO5
20. Evaluate the role of PHP in handling form data and the security considerations associated with form handling.
Discuss the differences between PHP 5 and PHP 7, emphasizing the performance improvements and new features.
CO6, PO6
21. Evaluate the security features provided by ASP.NET and how they contribute to building secure web applications.
Compare and contrast the security considerations in PHP and ASP.NET, highlighting best practices for securing web applications in both environments.
CO6, PO6